



text classification & Naive Bayes

LING 430 Computational Linguistics Fall 2026

lecture 7b

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last time

- natural language generation
 - decoding: greedy and sampling
 - softmax and temperature
 - top-k & top-p



agenda

- text classification
- Naive Bayes
- evaluation



today's notebook

class07b.ipynb

- course website → week 7 → activity
 - **File → Save a copy in Drive**



text classification



is this spam?

Subject: Fwd: If your schedule permits, I would love to secure a brief slot on your calendar

Hi Pat,

I just got this email with your name, and I'm wondering if it's spam...?

----- Forwarded message -----

From: **Patricia DeLucia** <dunnchristophervin@gmail.com>

Date: Mon, Aug 31, 2026 at 2:08 PM

Subject: If your schedule permits, I would love to secure a brief slot on your calendar

To: siyu.liang@rice.edu <siyu.liang@rice.edu>

Patricia DeLucia

Associate Dean for Research, School of Social Sciences

Professor, Psychological Sciences

Member, Ken Kennedy Institute

Member, Institute of Health Resilience and Innovation

CONTACT

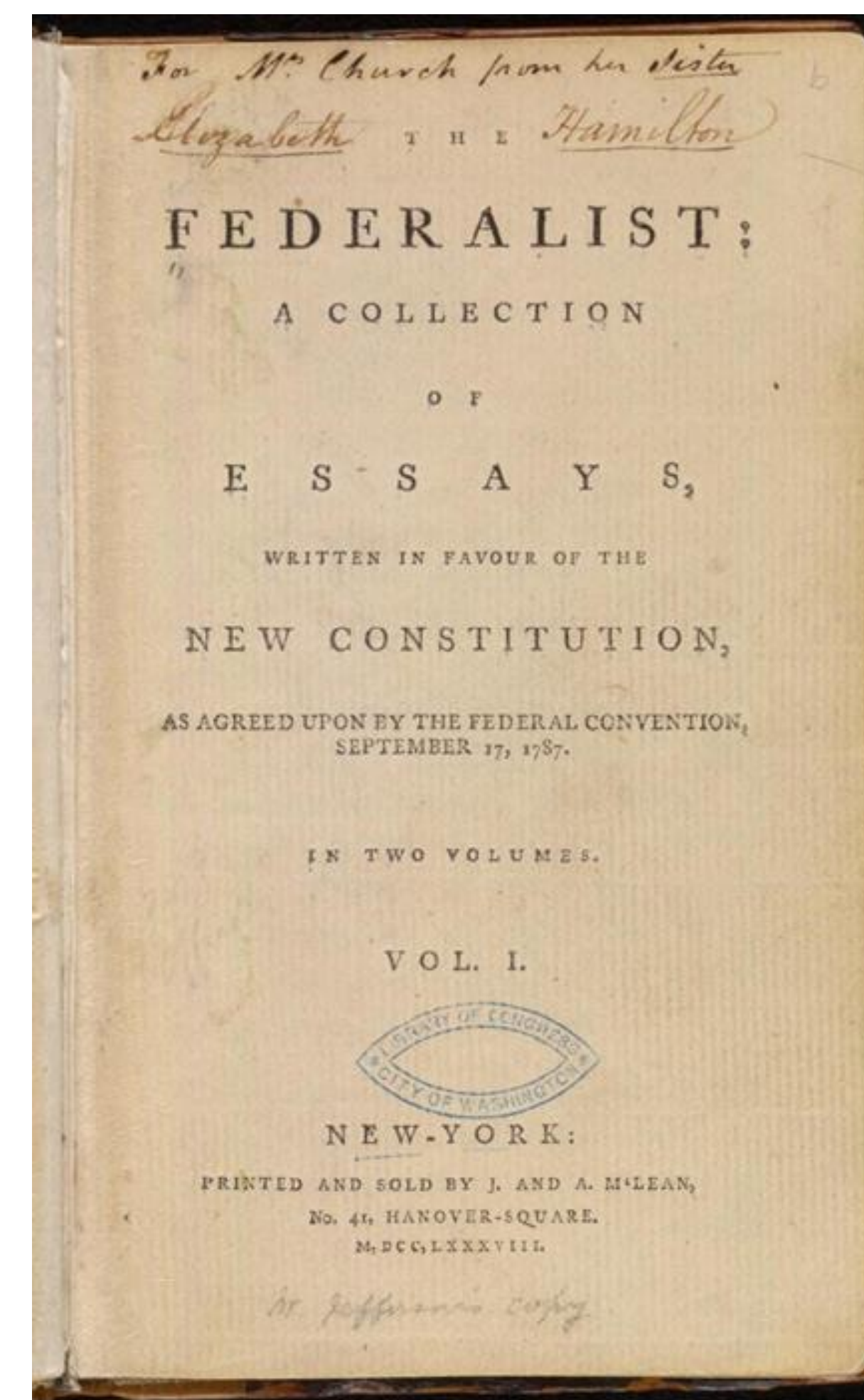
Sewall Hall 492A

School of Social Sciences

Rice University

who wrote which Federalist Papers?

- 1787–88: 85 essays published anonymously by Alexander Hamilton, James Madison and John Jay
- the authorship of 12 of them was disputed between Hamilton and Madison
- 1963: settled by Mosteller and Wallace, counting function words such as *upon* and *whilst*, with Bayes' theorem



positive or negative?

- *unbelievably disappointing* 👎
- *full of zany characters and richly applied satire, and some great plot twists* 👍
- *this is the greatest screwball comedy ever filmed* 👍
- *it was pathetic; the worst part about it was the boxing scenes* 📉

what is the subject of this article?

- linguistics?
- animal science?

Toward Automatic Discovery of a Canine Phonetic Alphabet

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Abstract

Dogs communicate intelligently through vocalizations, yet the phonetic properties of their communication remain poorly understood. This paper introduces an iterative algorithm inspired by human phonetic discovery, leveraging minimal pairs to identify distinct *canine cognitive vocal units* and construct a comprehensive phonetic alphabet. Additionally, the algorithm derives *canine vocal pattern* that exhibit structured correlations with specific environments and activities, suggesting potential meaningful communicative patterns. Our approach provides a novel framework for analyzing non-human vocalization systems and offers potential applications beyond canines to other animal species, advancing the study of cross-species communication.

1 Introduction

Animals have a variety of ways of communicating

Recent advancements in machine learning have broadened the path of “animal language” research, offering the possibility of a data-driven study that can be conducted at a more granular level. [Huang et al. \(2023\)](#) applied human International Phonetic Alphabet (IPA) directly to transcribe Shiba Inu voices (Figure 1c), which restricts dog sounds to a subset of human IPAs, ignoring the differences between human pronunciations and dog vocals. [Hagiwara et al. \(2024\)](#) proposed Inter-Species Phonetic Alphabet (ISPA), a system designed to transcribe animal sounds into discrete symbols. However, the system was trained on datasets from hundreds of many different species (containing marine mammal, birds, dogs, and even mosquitos) to obtain non-human phones (Figure 1a). This coarse-grained approach overlooks the differences in vocal characteristics across species. [Wang et al. \(2024\)](#) used HuBERT ([Hsu et al., 2021](#)) to obtain “phonemes” in Shiba Inu, but many phonemes thus discovered

text classification

- **classification**: assigning each input one label from a fixed set
- today we'll talk about **text classification**
 - assigning subject categories, topics, or genres
 - spam detection
 - authorship identification (who wrote this?)
 - language Identification (is this Portuguese?)
 - sentiment analysis
 - ...



text classification

definition

- input
 - a document d
 - a fixed set of class labels $C = \{c_1, c_2, \dots\}$
- output
 - one predicted class label $c \in C$
- the clues the decision rests on are **features**
 - for our case, it would be just the words w

text classification

hand-coded rules

- a basic method is using hand-coded rules based on combinations of words or other features
- black-list-address OR (“dollars” AND “have been selected”) → spam
- accuracy can be high
 - in very specific domains
 - if rules are carefully refined by experts
 - but
 - building and maintaining rules is expensive
 - they are too literal and specific: “high-precision, low-recall”

text classification

supervised learning

- we can also use **supervised learning**
 - we need a **training set** of documents already labeled by people, i.e. the gold standard
 - a model learns what each label looks like, then labels new documents on its own

text classification

supervised learning

- we need
 - a document d
 - a fixed set of labels C
 - a training set of m labeled documents $(d_1, c_1), \dots, (d_m, c_m)$
- we will build
 - a classifier γ that maps a document to a label $\gamma(d) = c$

text classification

supervised learning

- there are many kinds of classifiers!
 - Naive Bayes (this lecture)
 - logistic regression (next week)
 - neural networks (week 12)
 - k-nearest neighbors
 - ...
- we can also use pretrained large language models
 - fine-tuned as classifiers
 - prompted to give a classification



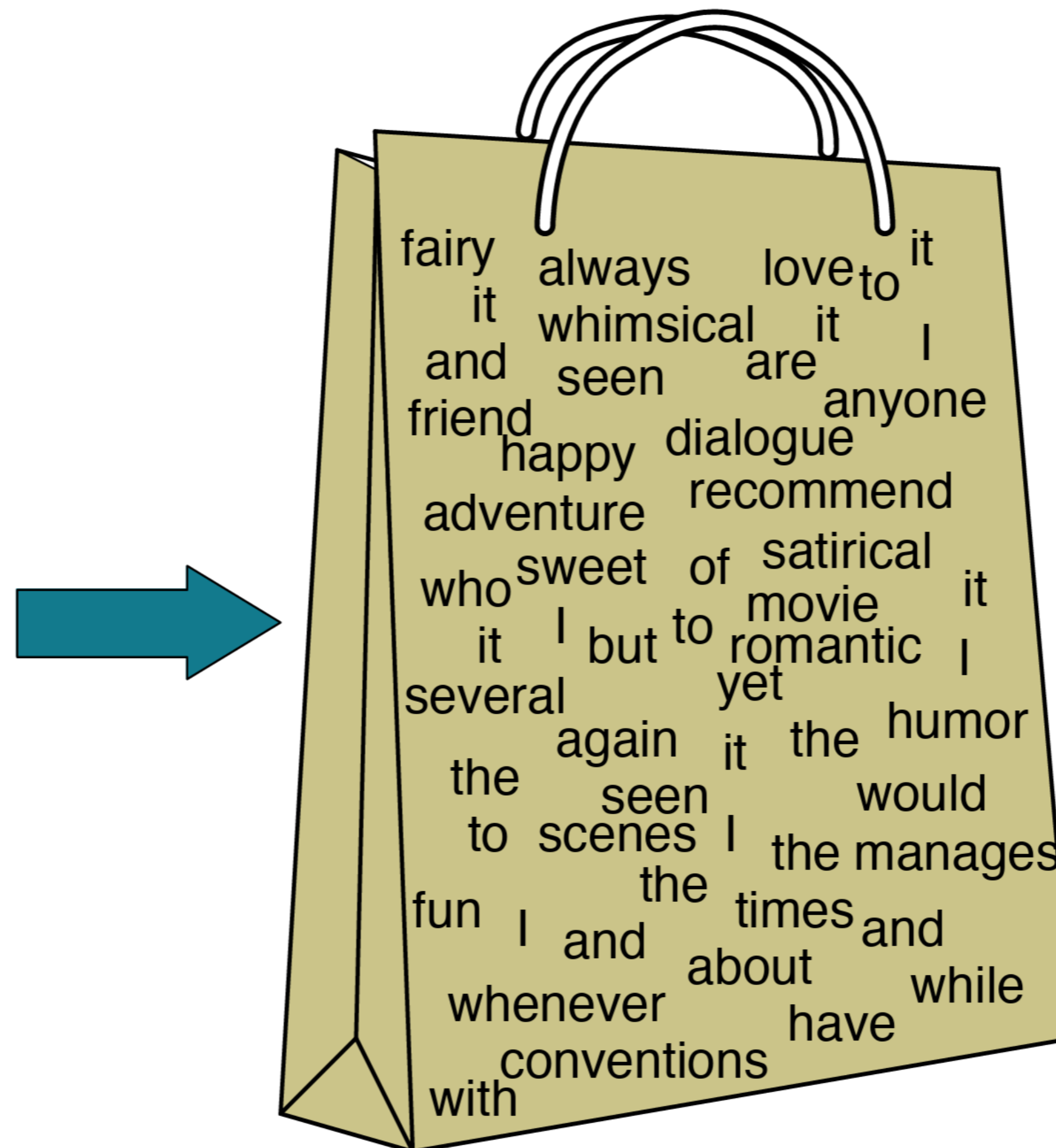
Naive Bayes

Naive Bayes

- it's a simple (“Naive ”) classification method based on the Bayes theorem
- relies on very simple assumption of document, namely **bag of words**

bag of words

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

bag of words

$$\gamma \left(\begin{array}{cc} \text{seen} & 2 \\ \text{sweet} & 1 \\ \text{whimsical} & 1 \\ \text{recommend} & 1 \\ \text{happy} & 1 \\ \dots & \dots \end{array} \right) = C$$

 / 

Naive Bayes

- what we want to do
 - for a document d , choose the label c with the highest probability given the document

$$c = \operatorname{argmax}_c P(c|d)$$

- but we don't know $P(c|d)$
 - the model hasn't seen this review d during training
- Bayes' theorem turns it around!

Naive Bayes

- Bayes' theorem for a document d and a class c

$$P(c|d) = \frac{P(d|c) \times P(c)}{P(d)}$$

- we want to calculate

$$c = \operatorname{argmax}_c P(c|d)$$

- apply the Bayes' theorem to our question

$$c = \operatorname{argmax}_c \frac{P(d|c) \times P(c)}{P(d)}$$

- $P(d)$ is the same for every label, so we drop it

$$c = \operatorname{argmax}_c P(d|c) \times P(c)$$

Naive Bayes

- now we have

“likelihood”

“prior”

$$c = \operatorname{argmax}_c P(d|c) \times P(c)$$

- we know $P(c)$, but how do we calculate $P(d|c)$?

Naive Bayes

the “Naive” assumption

- $P(d|c)$ is the probability of a review text d with label c
- a text is a sequence of words
- two simplifications
 - **bag of words**: the position of a word does not matter
 - **conditional independence**: given the label, each word is independent of the others

$$P(d|c) = P(w_1, \dots, w_n|c) = P(w_1|c) \times P(w_2|c) \times \dots \times P(w_n|c)$$

- “Naive” because words are **not** independent
 - after *very*, *good* is far likelier than *film*
 - but the assumption gives us something that works

Naive Bayes

- now putting all the pieces together

$$\begin{aligned} c &= \operatorname{argmax}_c P(w_1, \dots, w_2 | c) \times P(c) \\ &= \operatorname{argmax}_c P(c) \prod_i P(w_i | c) \end{aligned}$$

- the “prior” $P(c)$: how common the label is
- the “likelihood” $P(w|c)$ of every word in the document
- the product is largest for the label that best explains the words

Naive Bayes

- now how do we estimate $P(c)$ and $P(w|c)$?
 - count in the training set
 - N_c reviews with label c , out of N training reviews

$$P(c) = \frac{N_c}{N}$$

- how often w occurs among all the words of the reviews with label c

$$P(w|c) = \frac{\text{count}(w, c)}{\sum_{w'} \text{count}(w', c)}$$

Naive Bayes

unknown words

- what if a word never occurs with a label?
 - new words, language change, typos, ...
- suppose no positive training review contains *fantastic*

$$P(\text{fantastic}|\text{pos}) = \frac{0}{\sum_{w'} \text{count}(w', \text{pos})} = 0$$

- one zero makes the whole product zero! 

Naive Bayes

unknown words

- we can ignore them
 - remove them from the test document
 - pretend they weren't there!
 - don't include any probability for them at all!
- why don't we build an unknown word model?
 - it doesn't help
 - knowing which class has more unknown words is not generally helpful

smoothing

add-one smoothing

- add one (Laplace) to every count, and $|V|$, the vocabulary size, to the total

$$P(w|c) = \frac{\text{count}(w, c) + 1}{\sum_{w'} \text{count}(w', c) + |V|}$$

- every word of the vocabulary now has a probability above zero under every label

Naive Bayes

stop words

- stop words: very frequent words like the and a
- some systems ignore stop words
 - sort the vocabulary by word frequency in training set
 - call the top 10 or 50 words the stopword list
 - remove all stop words from both training and test sets as if they were never there
- but removing stop words doesn't usually help
- so, in practice, most NB algorithms use all words and don't use stop word lists

Naive Bayes classifier example

- you're training a model based on five reviews
- and then you're asked to predict the label of a test review

	label	review
training	neg	just plain boring
	neg	entirely predictable and lacks energy
	neg	no surprises and very few laughs
	pos	very powerful
	pos	the most fun film of the summer
test	?	predictable with no fun

Naive Bayes classifier example

1. the priors

- 3 of the 5 training reviews are negative

$$P(\text{neg}) = \frac{3}{5}$$

$$P(\text{pos}) = \frac{2}{5}$$

	label	review
training	neg	just plain boring
	neg	entirely predictable and lacks energy
	neg	no surprises and very few laughs
	pos	very powerful
	pos	the most fun film of the summer
test	?	predictable with no fun

Naive Bayes classifier example

2. drop unknown words

- *with* occurs in no training review
- it is left out of the test review
- only three words remain
 - *predictable, no, fun*

	label	review
training	neg	just plain boring
	neg	entirely predictable and lacks energy
	neg	no surprises and very few laughs
	pos	very powerful
	pos	the most fun film of the summer
test	?	predictable with no fun

Naive Bayes classifier example

3. the likelihoods

- negative reviews word count 14; positive reviews word count 9; $|V| = 20$; apply smoothing

$$P(\text{predictable}|\text{neg}) = \frac{1 + 1}{14 + 20}$$

$$P(\text{no}|\text{neg}) = \frac{1 + 1}{14 + 20}$$

$$P(\text{fun}|\text{neg}) = \frac{0 + 1}{14 + 20}$$

$$P(\text{predictable}|\text{pos}) = \frac{0 + 1}{9 + 20}$$

$$P(\text{no}|\text{pos}) = \frac{0 + 1}{9 + 20}$$

$$P(\text{fun}|\text{pos}) = \frac{1 + 1}{9 + 20}$$

	label	review
training	neg	just plain boring
	neg	entirely predictable and lacks energy
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	pos	very powerful
	pos	the most fun film of the summer
test	?	predictable with no fun

Naive Bayes classifier example

4. scoring the test review

- multiply the prior with the three likelihoods, for each word in the set

$$P(\text{neg}) \times P(S|\text{neg}) = \frac{3}{5} \times \frac{2 \times 2 \times 1}{34^3} = 6.1 \times 10^{-5}$$

$$P(\text{pos}) \times P(S|\text{pos}) = \frac{2}{5} \times \frac{1 \times 1 \times 2}{29^3} = 3.2 \times 10^{-5}$$

- the test review is classified as negative

activity 1

- predict the label of a new test review based on the same training data
 1. find the prior
 2. find the the likelihood of each word under pos/neg
 3. get the product
- which label wins?

	label	review
training	neg	just plain boring
	neg	entirely predictable and lacks energy
	neg	no surprises and very few laughs
	pos	very powerful
	pos	the most fun film of the summer
test	?	very fun film

Naive Bayes

overflowing

- a real review can have hundreds of words
 $0.0006 \times 0.0007 \times 0.0009 \times 0.01 \times 0.5 \times 0.000008 \times \dots$
- the product underflows to 0 in a computer
- use logs: $\log(a \times b) = \log(a) + \log(b)$, so sum log probabilities instead
$$c = \operatorname{argmax}_c \log P(c) + \sum_i \log P(w_i | c)$$
- the label with the largest sum has the largest product
- still fits the same data

Naive Bayes

relationship to language model

- $P(w|pos)$ over the vocabulary is a unigram language model of positive reviews
 - similar for $P(w|neg)$
- a test review gets a probability from each; the higher one wins
 - $P(s|pos) > P(s|neg)$

word	$P(w pos)$	$P(w neg)$
i	0.1	0.2
love	0.1	0.001
this	0.01	0.01
fun	0.05	0.005
film	0.1	0.1

Naive Bayes in other tasks

- spam filtering: features beyond words
 - the subject line is all capitals
 - the sender address starts with many numbers
 - *one hundred percent guaranteed*
- language identification: character n-grams as features, *th*, *sch*, *ng*
 - train on many varieties of each language, or the classifier learns one dialect as the language

Naive Bayes is not so naive

- fast, and small: two count tables
- works with very little training data
- robust to irrelevant words: they push both labels equally and cancel out
- a dependable baseline
 - logistic regression, for example, does better by learning how much each word matters



evaluation

activity 2

- run the notebook today
- with a neighbor, read five of each and think about what tripped the classifier
- are the two kinds of error (false positive and false negative) tripped by the same things?

HW4

- released today, due **Tuesday, Oct 20, 11:59 pm**
 - content covered: n-gram, probability, natural language generation, Naive Bayes classification, evaluation
 - you get longer time to work on it due to midterm recess
 - happy studying/traveling!



next time

- Tuesday, Oct 13: no class, midterm recess
- Thursday, Oct 15
 - logistic regression